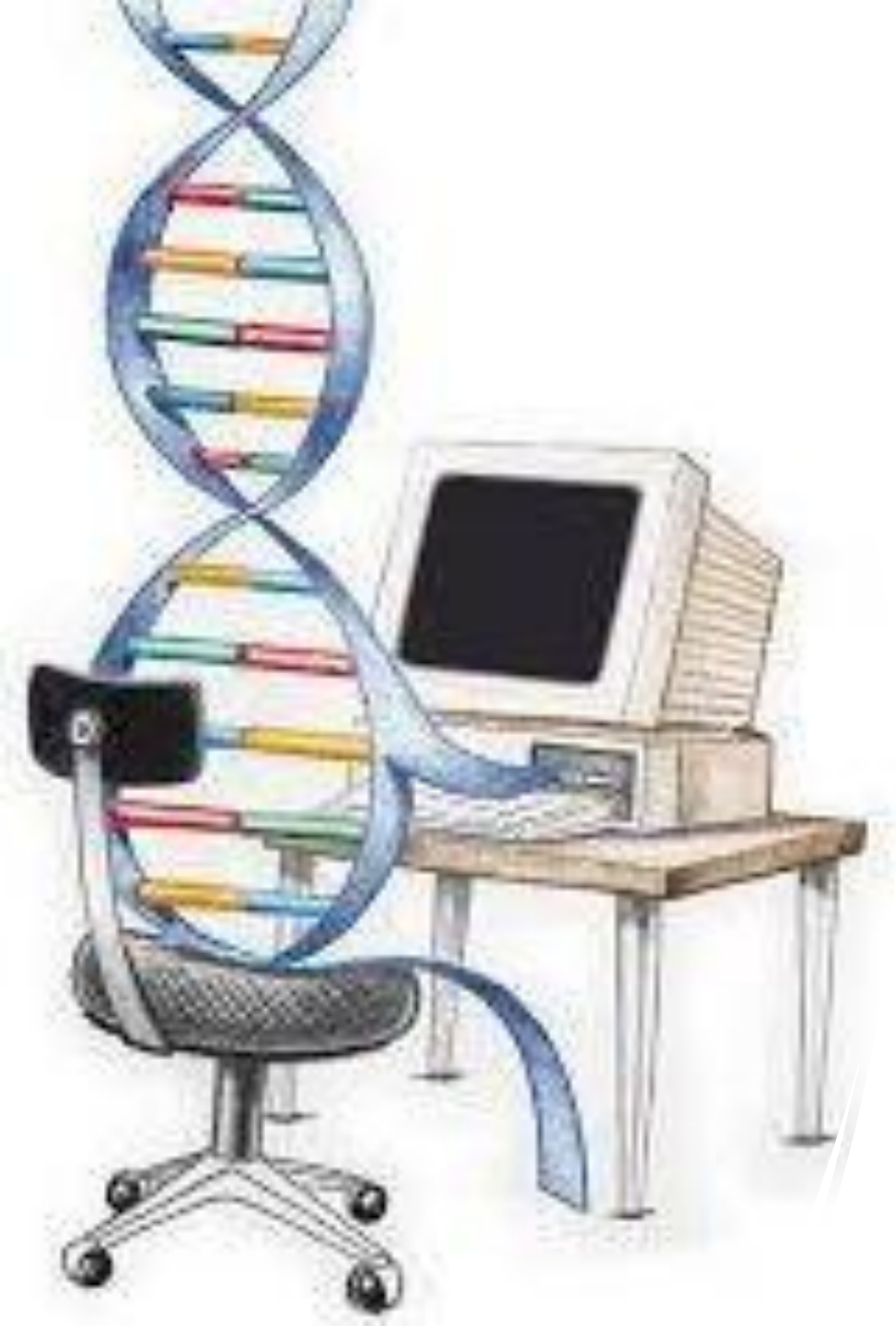


# Introduction to Evolutionary Computation
# **Genetic Algorithms**

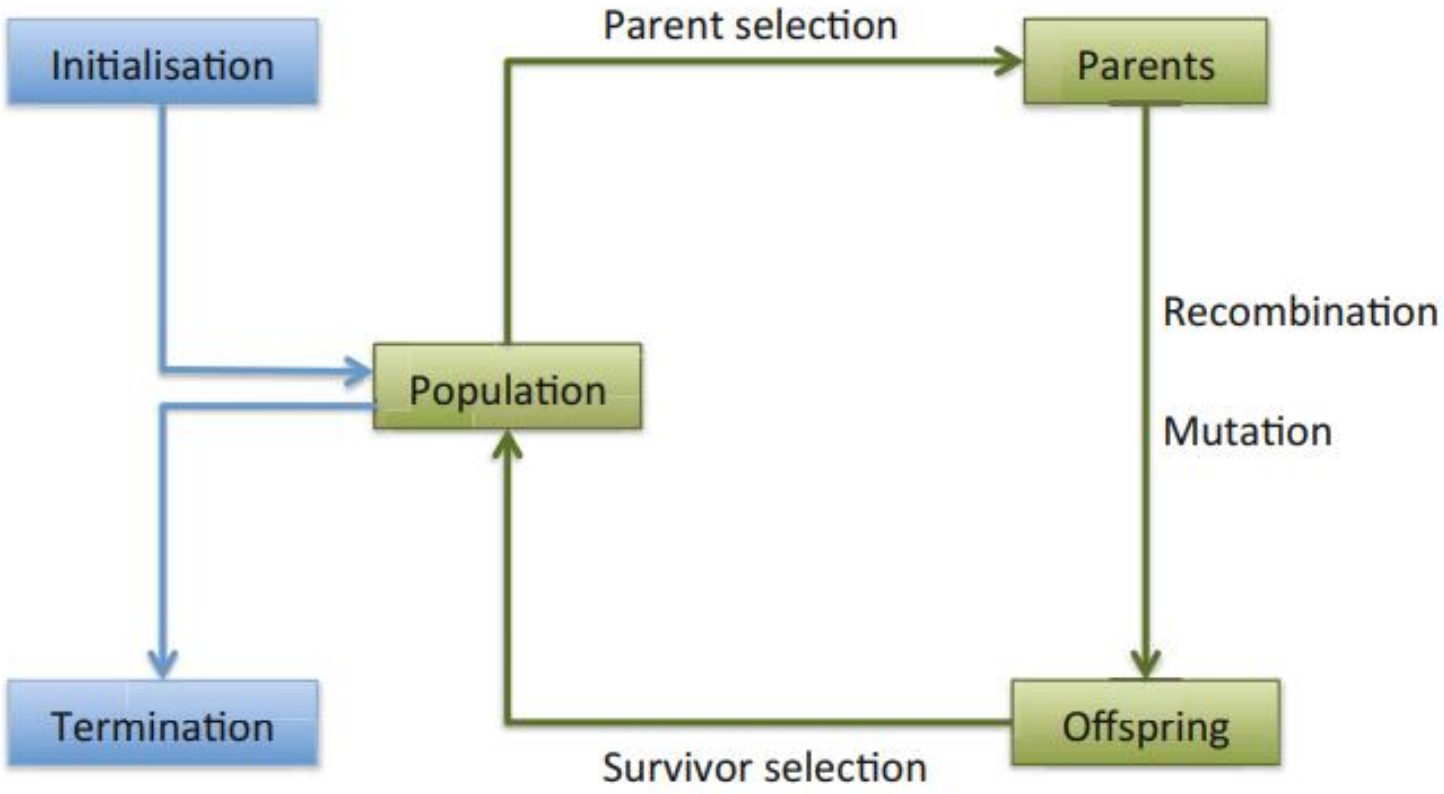
# Evolution- Racehorse Bloodlines

- Racehorse competition
- **Goal:** Run FAST!

- **Solution:** Champion Horses (DNA)
- **Optimization :** Breeding history
- Racehorse's origin from 3 horses
  - Byerley Turk (1680-1703 AD)
  - Darley Arabian (1700-1722 AD)
  - Godolphin Arabian (1724-1753 AD)

- Select good horses (winners & relatives)
- Breed
- next generation of horses
- competitions

# The general scheme of an evolutionary algorithm as a flowchart



# Genetic Algorithms

- John Holland, 1960s
- Based on ideas from Darwinian Evolution
- Can be used to solve a variety of problems that are not easy to solve using other techniques

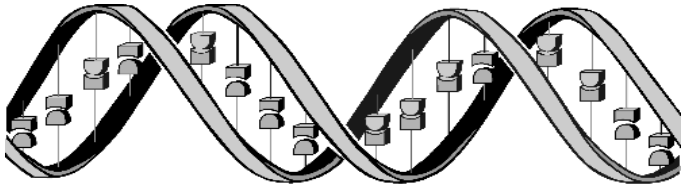
- “Adaptation in natural and artificial systems”, 1975.

# Principle of Natural Selection

- Giraffes with slightly longer necks could feed on leaves of higher branches when all lower ones had been eaten off.
- They had a better chance of survival.
- Favorable characteristic propagated through generations of giraffes.
- Now, evolved species has long necks.

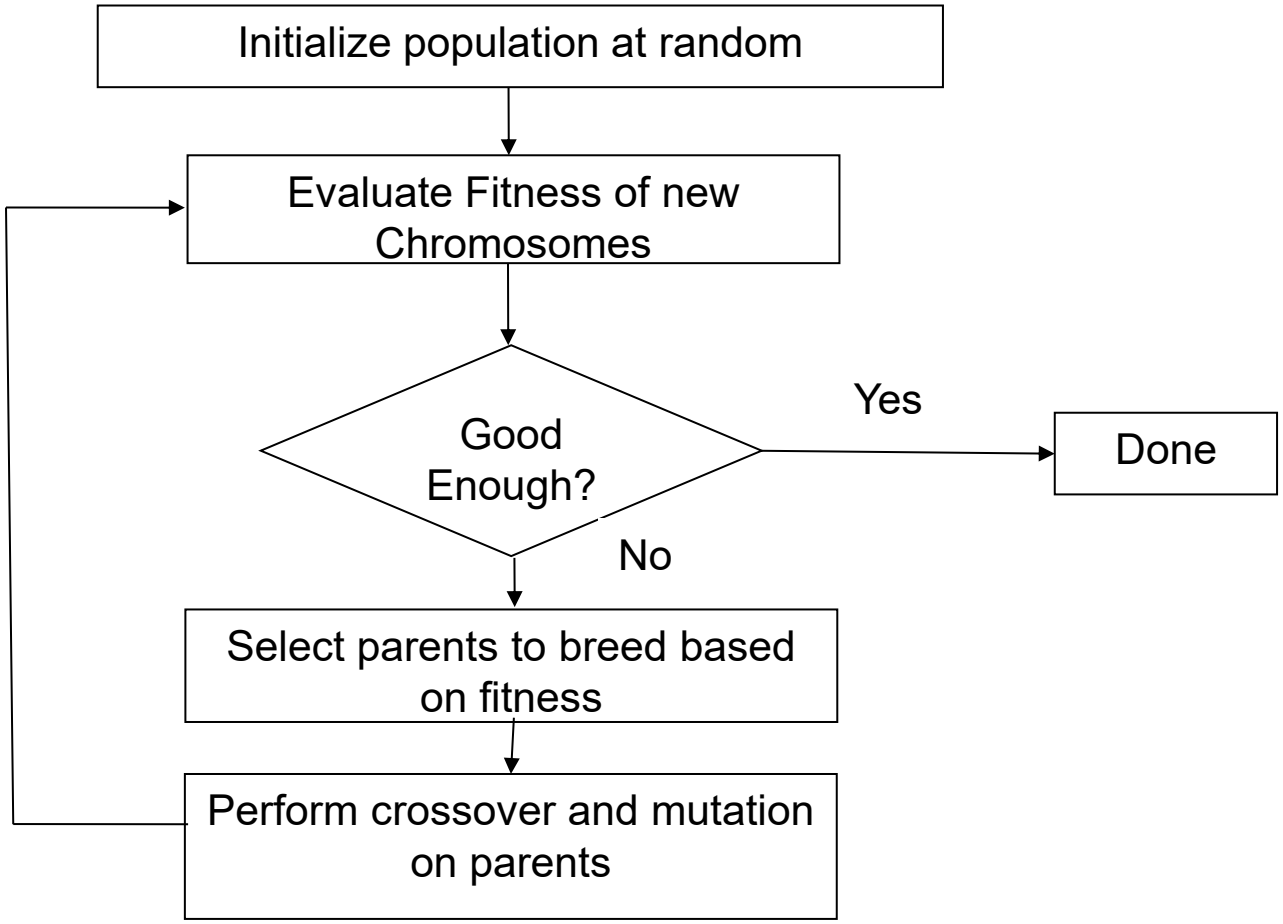
# Genetic Algorithms

- Problems are not solved through logical reasoning alone.
- Instead, populations of competing candidate solutions are created.
- These solutions evolve over generations, becoming progressively better.
- The evolution process mimics biological evolution:
  - **Selection:** Choosing the fittest candidates.
  - **Mutation:** Introducing random variations.
  - **Reproduction:** Combining traits of successful candidates.



- Survival of the Fittest:
  - Less effective candidates are eliminated.
  - Promising candidates survive and reproduce.
- This approach refines solutions iteratively toward optimal or near-optimal outcomes.

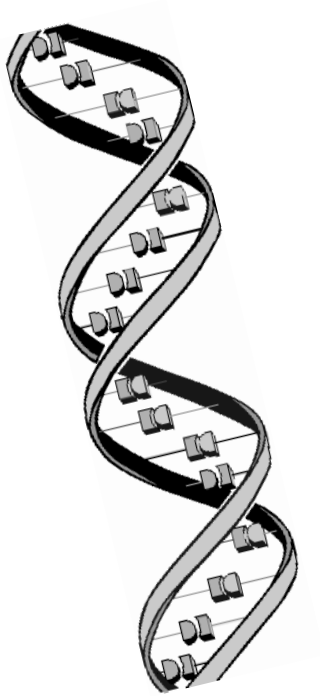
# Basic Flow of GA

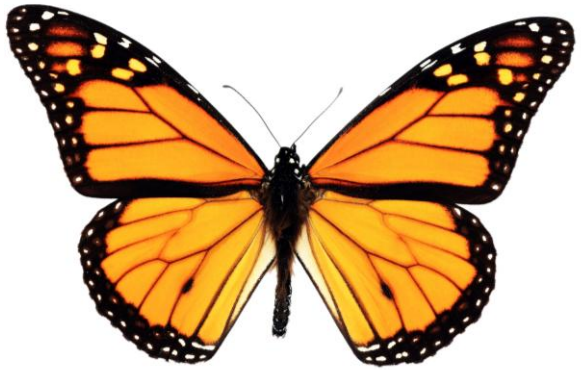


# Basics

- **Gene:** The basic unit representing a single characteristic or feature of an individual (like eye color). The specific value assigned to a gene is called an allele.
- **Chromosome:** A sequence of genes that collectively represents an individual or a possible solution to the problem. Each chromosome corresponds to a point in the search space.
- **Population:** A group of chromosomes that forms the entire set of candidate solutions in a given generation.
- **Importance of Representation:** Choosing an appropriate chromosome representation is crucial for the efficiency and complexity of the genetic algorithm (GA), as it impacts the algorithm's ability to explore and exploit the search space effectively.

# Genotype vs Phenotype





**Genotype**

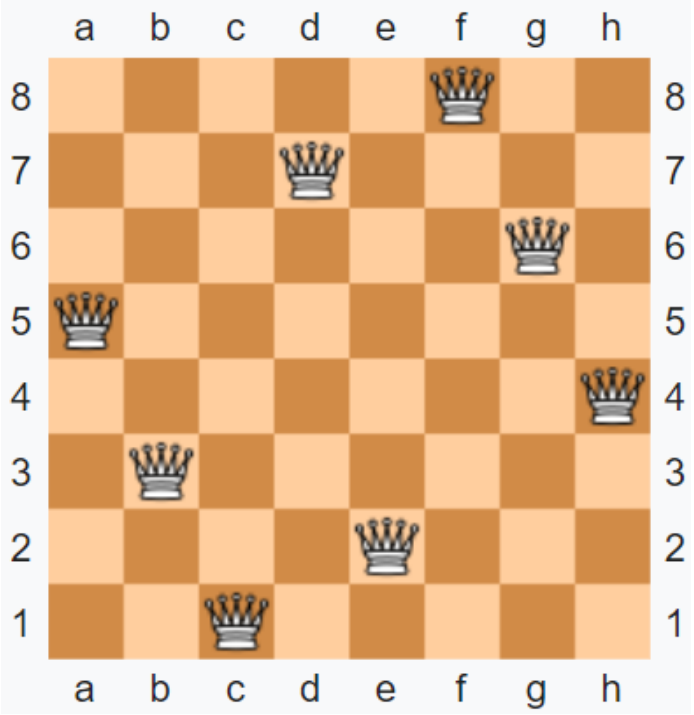
**Phenotype**

# The 8 Queen Problem: Representation

| 5 | 3 | 1 | 7 | 2 | 8 | 6 | 4 |
|---|---|---|---|---|---|---|---|

**Genotype: a permutation of the numbers 1 - 8**

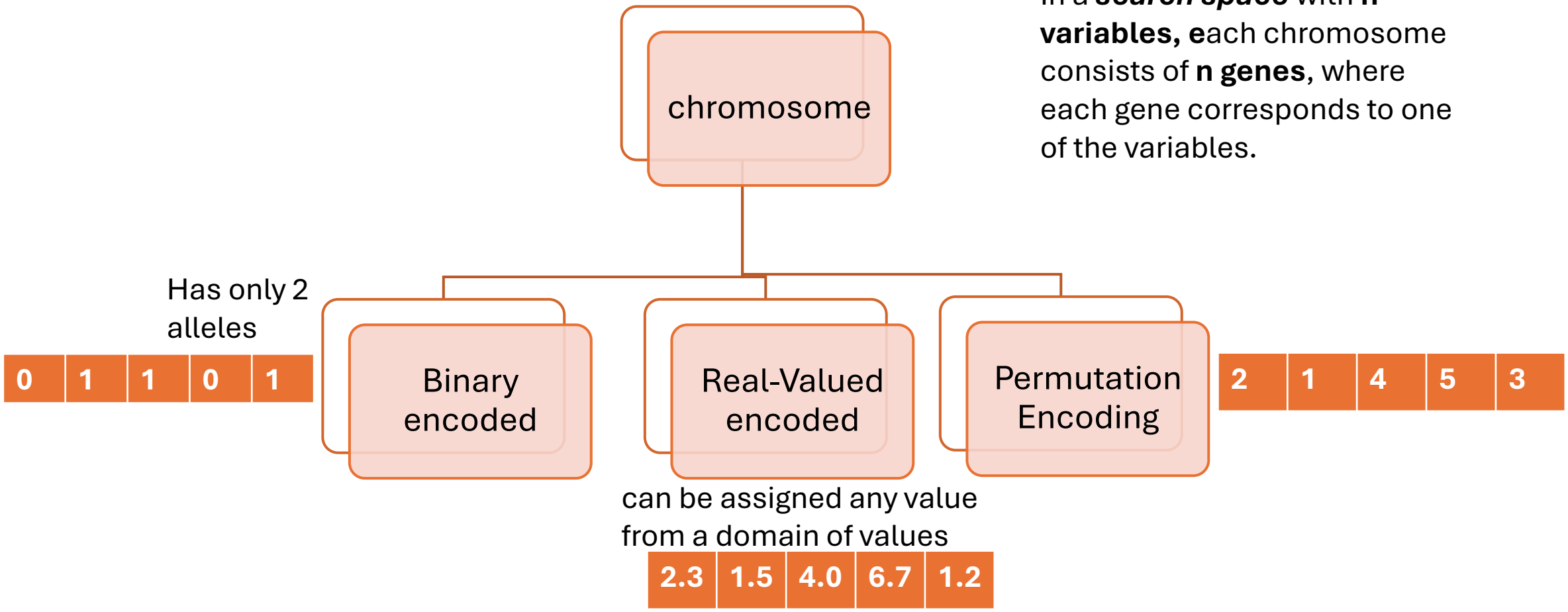
mapping




**Phenotype:** a board configuration

# Chromosome

- Chromosome is a vector of fixed length.

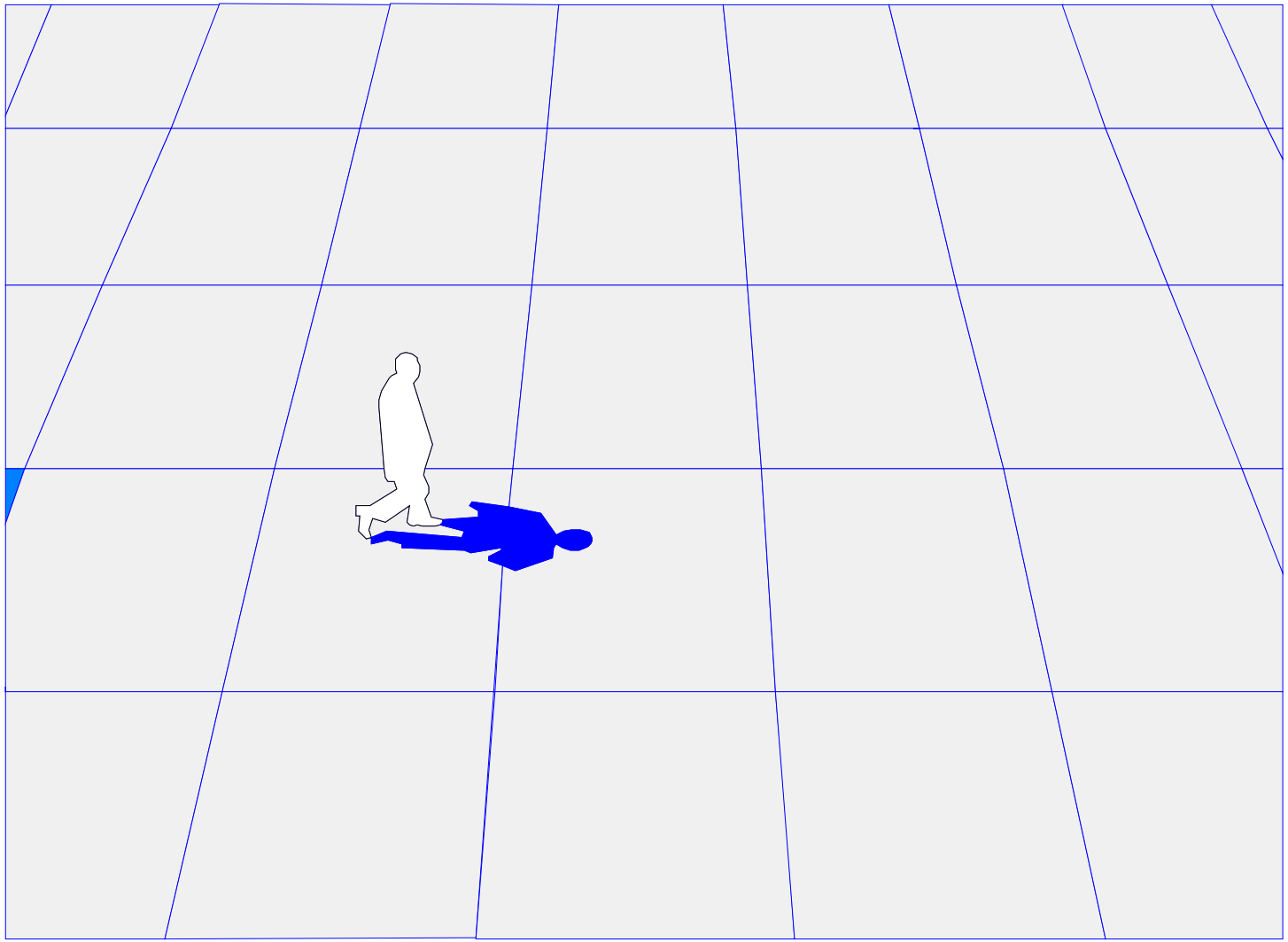



In a ***search space*** with **n variables, e**ach chromosome consists of **n genes**, where each gene corresponds to one of the variables.

# Genetic Algorithms

- *“Genetic Algorithms are good at taking large, potentially huge search spaces and navigating them, looking for optimal combinations of things, solutions you might not otherwise find in a lifetime.”*

- - Salvatore Mangano
- *Computer Design*, May 1995



# Chromosome Design for Personal Finance Management Problem

- Design a chromosome for Personal Finance Management Problem. You need to optimize how to allocate your monthly income to different expenses (e.g.,[1]rent, [2]groceries,[3]savings,[4]entertainment) while maximizing savings and minimizing overspending. Minimum you can allocated $10 and maximum $1000.

  - **Chromosome Representation:**
    - Each **gene** in the chromosome could represent the amount of money allocated to a specific expense.
    - For example, a chromosome could be represented as `[800, 200, 150, 50]` indicating allocations of $800 for rent, $200 for groceries, $150 for savings, and $50 for entertainment.

# Chromosome for Clothing Combination for the Day.

- Design a chromosome for Clothing Combination for the Day. You want to decide what clothes (a jacket, a hat, tie, gloves, and long shoes.) to wear today based on weather conditions, personal preferences, and style.

  - **Chromosome Representation:**
    - Each clothing item is represented by a **gene**, where **1** means the item is chosen to wear, and **0** means it is excluded.
    - For example, a chromosome `[1, 0, 1, 0, 1]` might represent wearing a jacket, skipping a hat, and including pants and shoes.

# Evaluation (Fitness Function)

- **Purpose:** Quantifies the optimality of a solution to rank it against others.
- **Measure of Performance:** Indicates how close the solution is to the desired output/Goal.

- **Evaluation Process:**
  - Decodes a chromosome representation.
  - Assigns a fitness measure to the solution.

- **Problem-Specific:** The fitness function depends on the specific problem being solved.

# Initial Population

- **Population Generation:**
  - The evolutionary process begins by generating an initial population.

- **Standard Method:**
  - Gene values are randomly selected to ensure uniform representation of the search space.

- **Heuristic Initialization:**
  - If prior knowledge about the problem is available, heuristics can bias the population toward good solutions.
  - **Risk:** Heuristics may lead to premature convergence to a local optimum, as not all parts of the search space are explored.

# Initial Population-Impact of Population Size

**Small Population:**

- Covers a small
- Requires mor
  has lowe

**Large Population:**

- Covers a
- Requires fewe
  increases time

**Mutation to Compensate:**

For small populations, increasing the mutation rate can force exploration of a larger search space.

# Selection Operators

- Individuals are chosen based on fitness to form an intermediate population for:
  - **Survival:** Some remain unchanged.
  - **Mutation or Replacement:** Some are modified or replaced.
  - **Crossover and Mutation:** Offspring replace parents after genetic operations.

# Selection Operators

- **Selection pressure**

- **High Selection Pressure:**
  - Strong focus on fitter individuals.
  - Faster convergence but increased risk of premature convergence to a local optimum.
- **Low Selection Pressure:**
  - Slower convergence.
  - Better exploration of the search space, reducing the risk of local optima.

# Selection Operators

- Different selection schemes
  - Tournament selection
  - Roulette wheel selection
  - Rank-based selection
  - Random selection

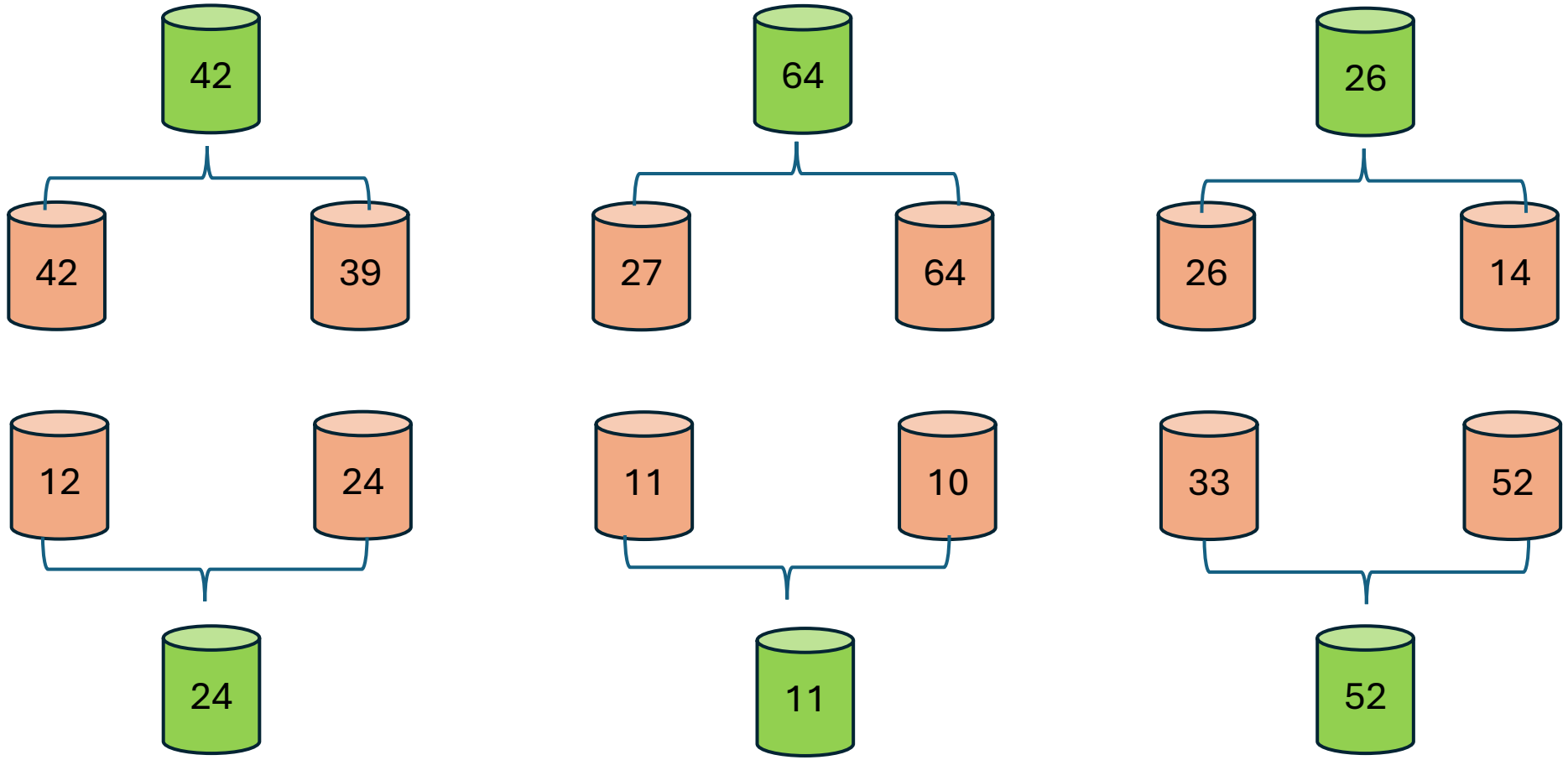
# Tournament selection

- **Process:**
  - Multiple tournaments are conducted among randomly chosen individuals from the population.
  - The winner of each tournament is selected for the next generation.

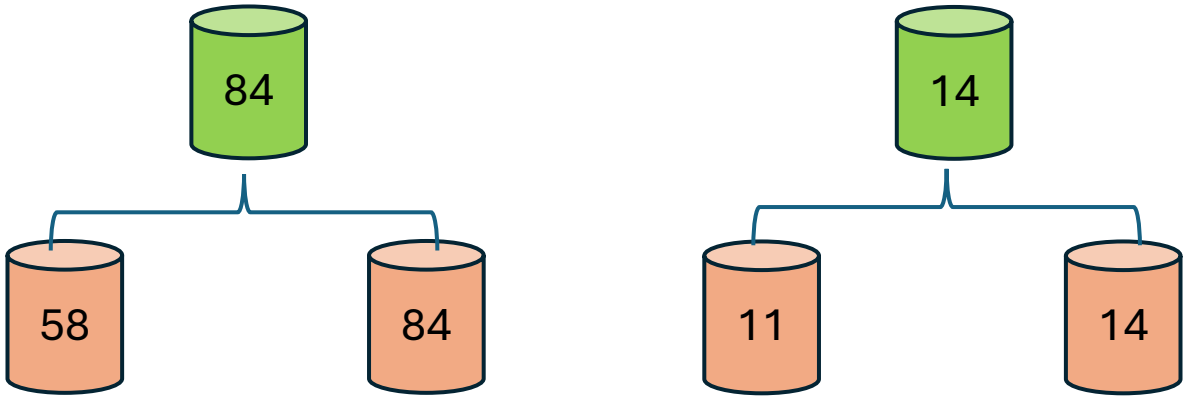
- **Adjusting Selection Pressure-Tournament Size:**
  - **Larger size** Favors stronger individuals, increasing selection pressure.

  - **Smaller size** Provides weaker individuals a higher chance of selection, reducing selection pressure.

# Tournament selection

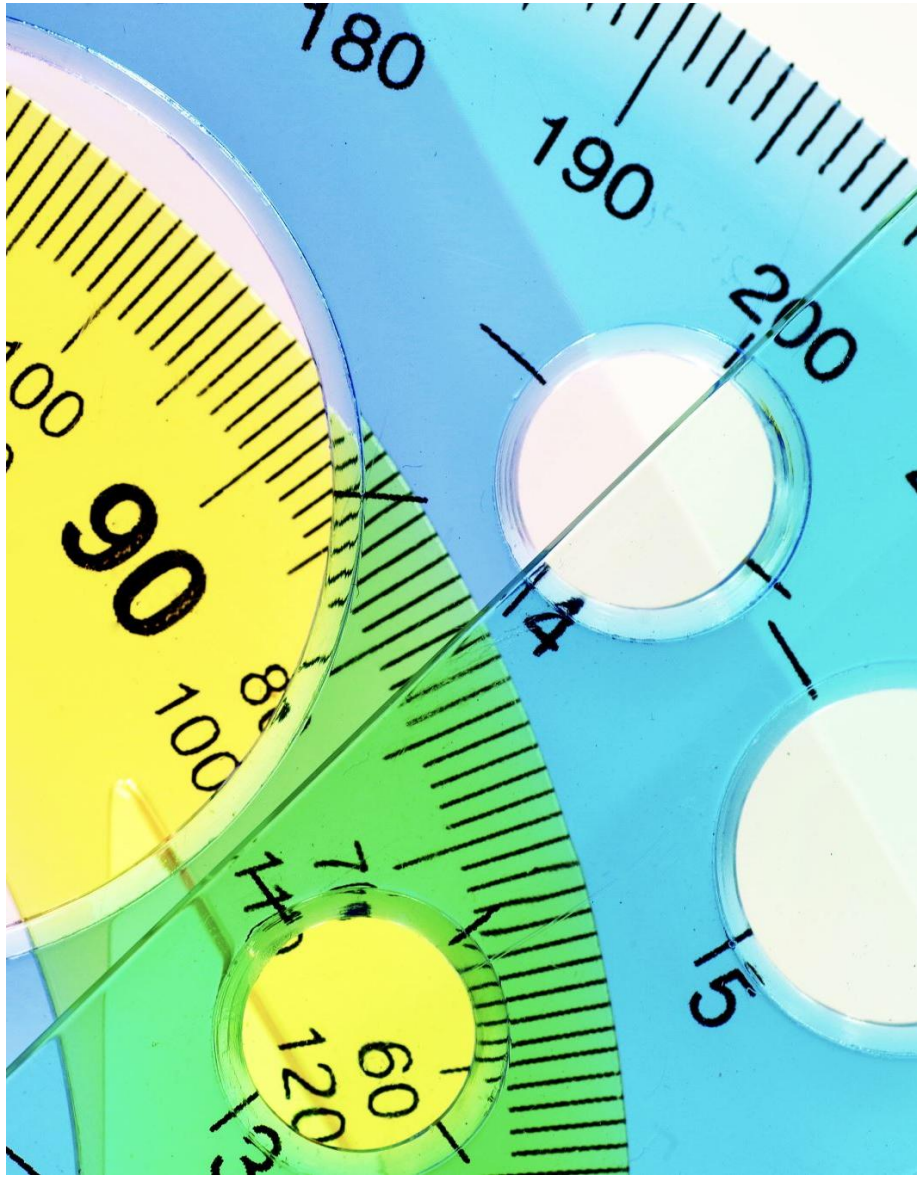


# Tournament selection



# Roulette wheel selection

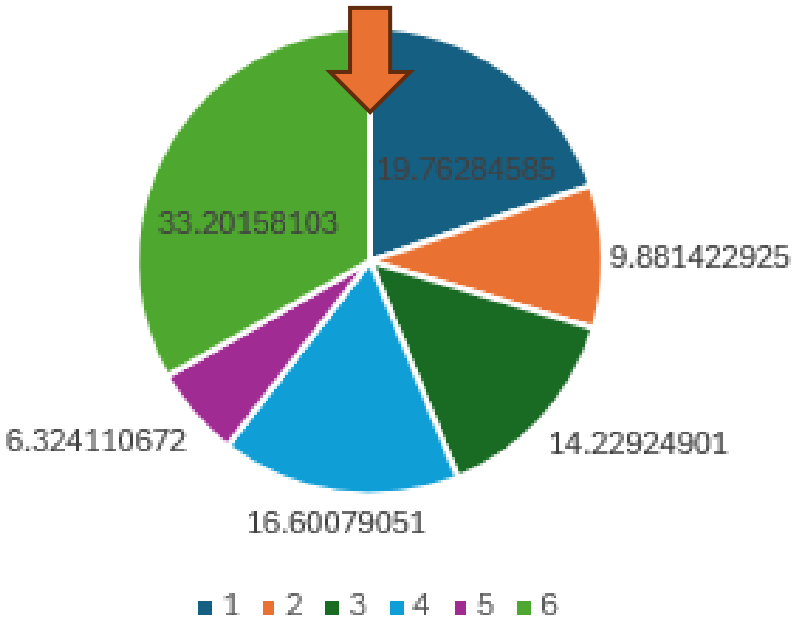
- Roulette Wheel Selection selects individuals from a population based on their fitness values, where those with higher fitness have a greater likelihood of being chosen.



# Roulette wheel /Proportionate selection

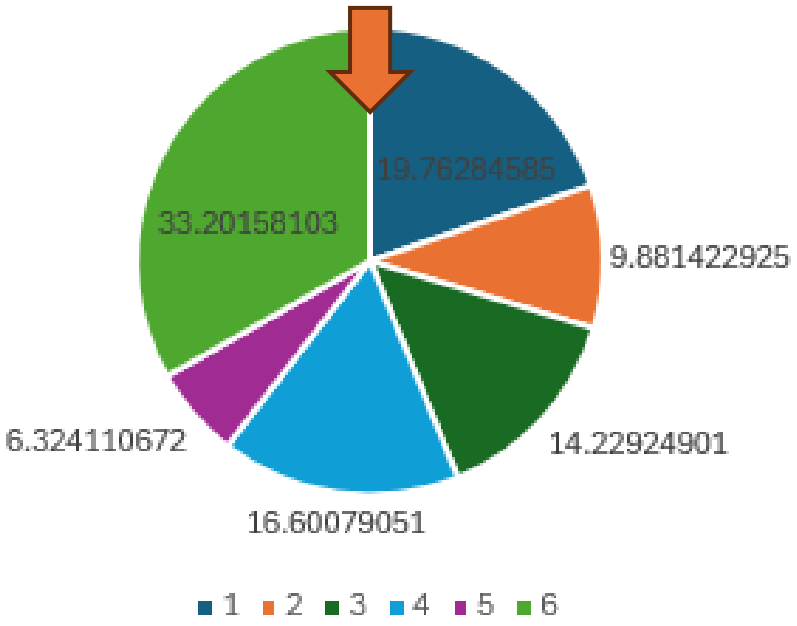
**Maximization Problem**

| Chromosome # | Fitness | % of RW | range |
|---|---|---|---|
| A | 50 | | |
| B | 25 | | |
| C | 36 | | |
| D | 42 | | |
| E | 16 | | |
| F | 84 | | |
| Total | | | |



# Roulette wheel /Proportionate selection

| Chromosome # | Fitness | % of RW | range |
|---|---|---|---|
| A | 50 | 19.76 | 0-19.76 |
| B | 25 | 9.88 | 19.76-29.64 |
| C | 36 | 14.23 | 29.64-43.87 |
| D | 42 | 16.60 | 43.87-60.47 |
| E | 16 | 6.32 | 60.47-66.8 |
| F | 84 | 33.20 | 66.8-100 |
| Total | 253 | 100 | |



# Rank-based selection-Method 1

- Rank-based selection assigns selection probabilities to individuals based on their rank rather than their raw fitness values.
- **Advantage**: This helps to avoid excessive selection pressure caused by large fitness value differences and ensures a more balanced selection process.

$$P_i = \frac{2 \times (\text{Rank of Individual})}{N \times (N+1)}$$

where $N$ is the total number of individuals.

# Rank-based selection

| Chromosome# | Fitness | Rank | Probabilities | Range |
|---|---|---|---|---|
| F | 84 | | | |
| A | 50 | | | |
| D | 42 | | | |
| C | 36 | | | |
| B | 25 | | | |
| E | 16 | | | |

# Rank-based selection

**Maximization Problem:**
**6 for highest rank**
**1 for lowest rank**

| Chromosome# | Fitness | Rank | Probabilities | Range |
|---|---|---|---|---|
| F | 84 | 6 | 0.29 | 0.77-1 |
| A | 50 | 5 | 0.24 | 0.53-0.77 |
| D | 42 | 4 | 0.19 | 0.34-0.53 |
| C | 36 | 3 | 0.14 | 0.2-0.34 |
| B | 25 | 2 | 0.10 | 0.05-0.14 |
| E | 16 | 1 | 0.05 | 0-0.05 |

# Random selection

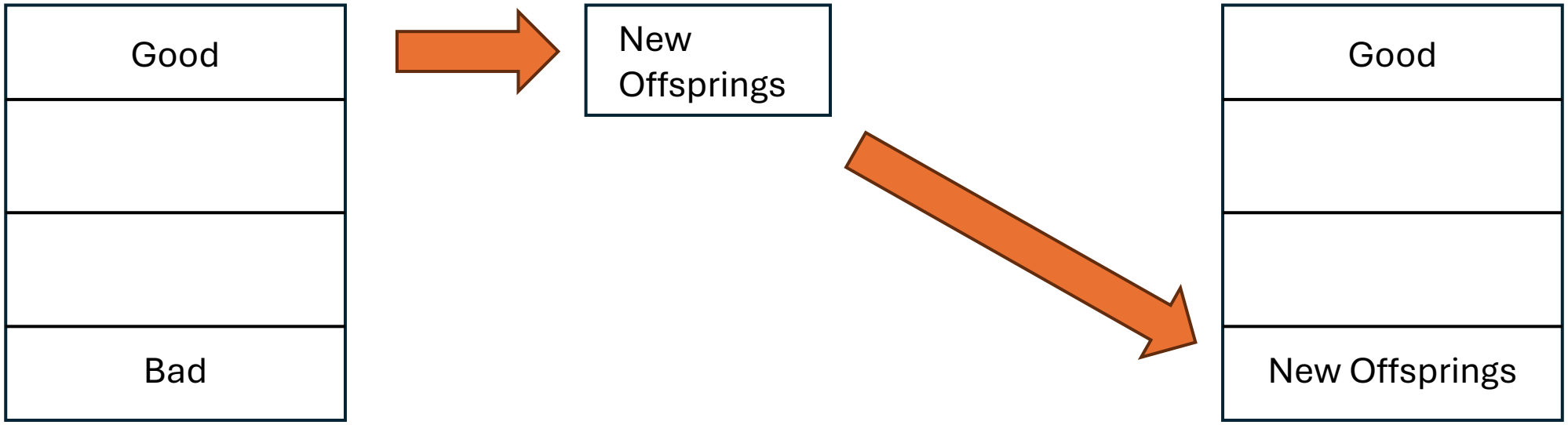
- Random selection selects individuals randomly from the population without considering their fitness values.
- Each individual has an equal chance of being selected, regardless of its quality as a solution.

# Steady state selection

In this method, a few good chromosomes are used for creating new offspring in every iteration.

Then some bad chromosomes are removed, and the new offspring is placed in their places

The rest of population migrates to the next generation without going through the selection process



# Generation Gap (GP)

- Number between 0 and 1
- It represents the fraction of population that gets replaced by offspring.
- If GP = 1, the new generation consists entirely of new individuals.
- For GP =0.4, 0.4*P offsprings are created and (1 - 0.4)*P individual from the last generation are preserved for the next generation.

# Elitism-Selection of k Individuals

- Elitism ensures that a fraction of best individuals from the current generation survives unchanged into the next generation.
- Number between 0 and 1
- If Pop_size = 20 and Elitism = 0.1 then 2 individuals are preserved and 18 may get replaced by offsprings. If GP = 0.9 then these 18 will definitely get replaced.

- Purpose:
  - To retain high-quality solutions across generations.
  - Prevents the loss of the best solutions due to stochastic variation.

# Exercise

- In genetic algorithm, a generation has 50 chromosomes, out of which 5 fittest ones are placed in the next generation without any competition from the offspring. The rest of the 45 chromosomes of the next generation are chosen from the pool of parents and offspring. Suppose 30 chromosomes of the offspring make it to the new generation. What is the generation gap and what is the elitism value. Duplicates are not allowed.

# Duplicates

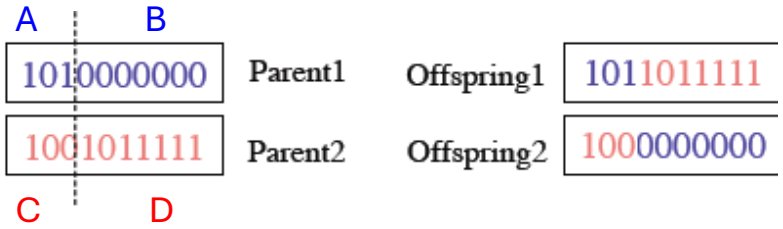
- Eliminating duplicates increases the efficiency of the genetic search and reduces the danger of premature convergence

- Eliminating duplicates means that if an offspring is created which is a duplicate of a chromosome of the current population, we terminate it immediately and create a new one. It increases the processing time in large populations

# Crossover Operator- Recombination

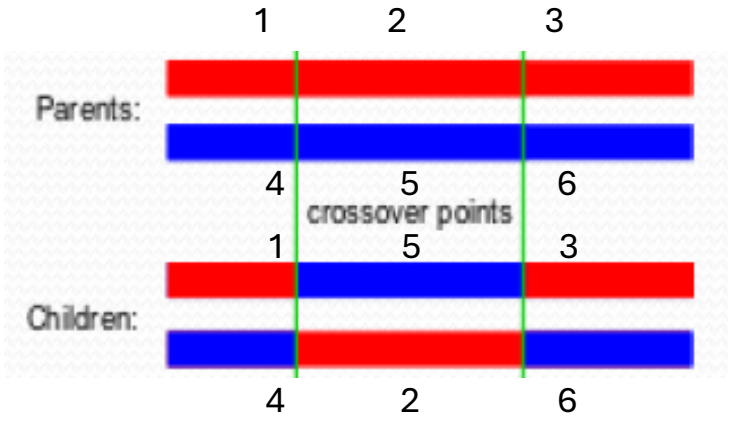
- Accelerates Early Search
- Effective Combination of Subsolutions
- Random Pair Selection
- Encoding-Dependent Crossover Methods
  - Examples include:
    - Single Point Crossover
    - Two Point Crossover

# Crossover Operator- Recombination

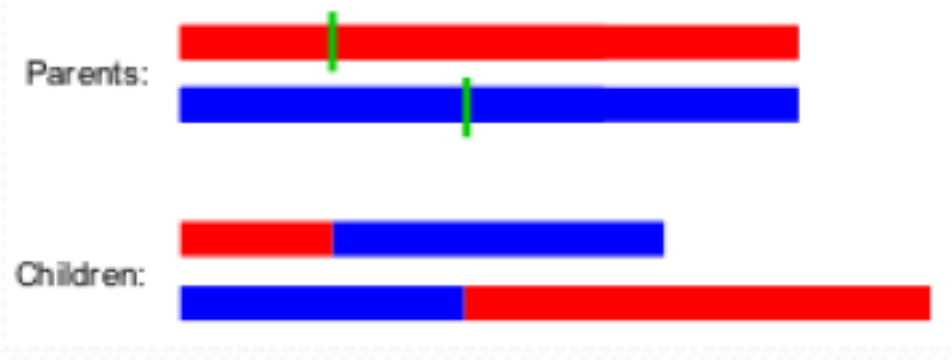




- Single Point Crossover

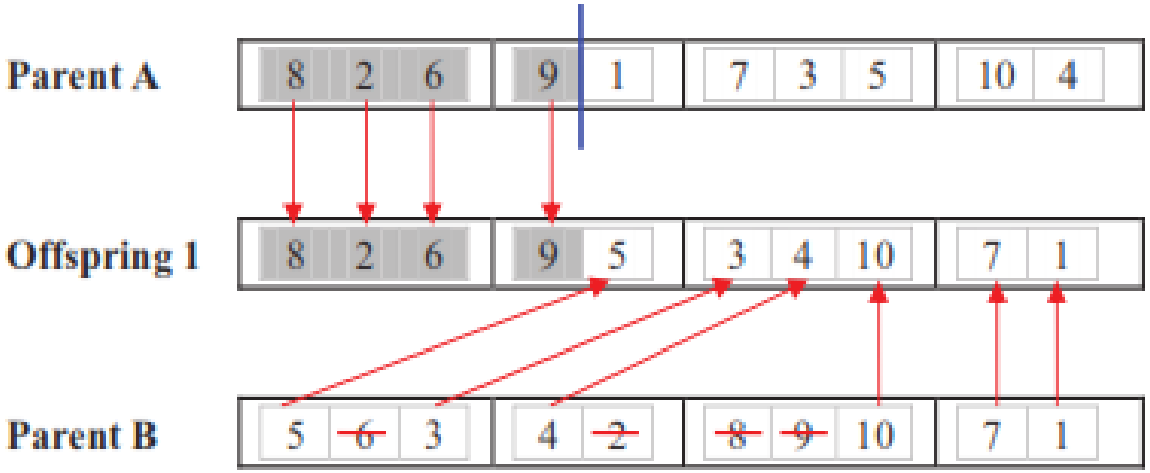



- Two Point Crossover

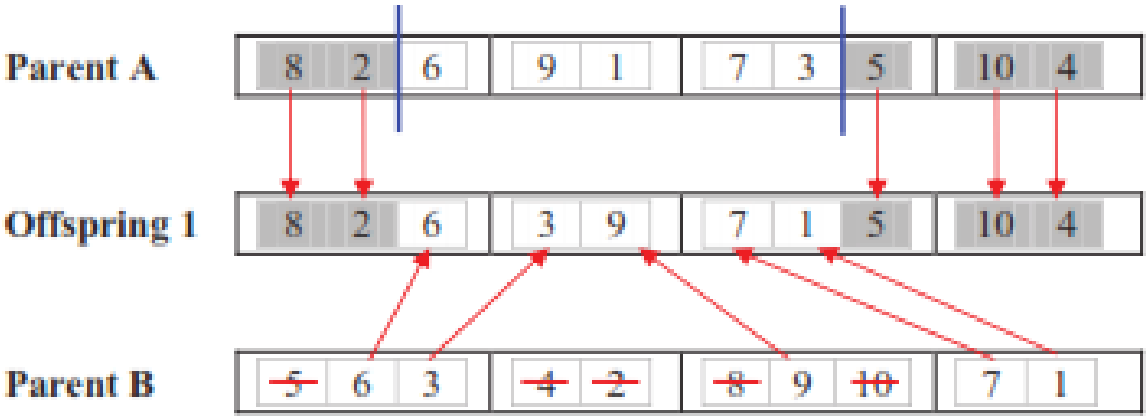



- Others like Cut and paste etc.

# Crossover operator




One-point crossover operator



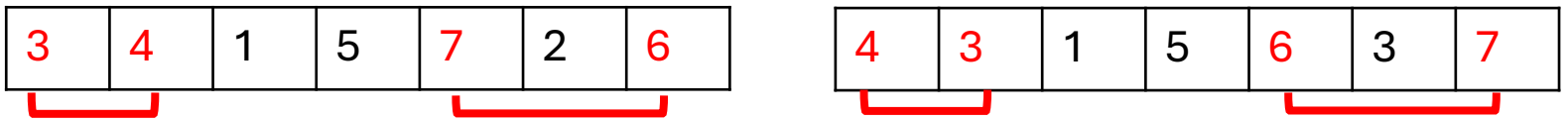

Two-point v1 crossover operator

# Mutation

- Bit Flip.


- Random Resetting (integers)


- Swap

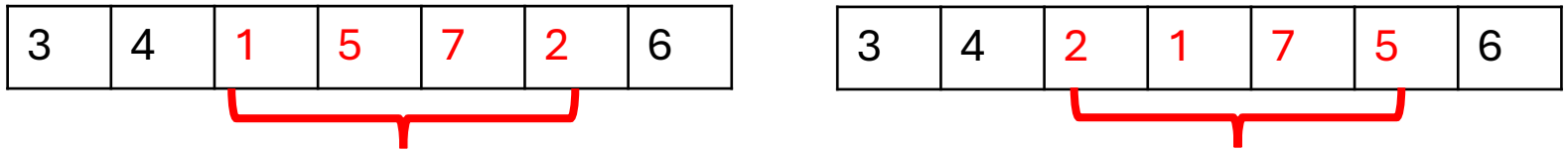


# Mutation

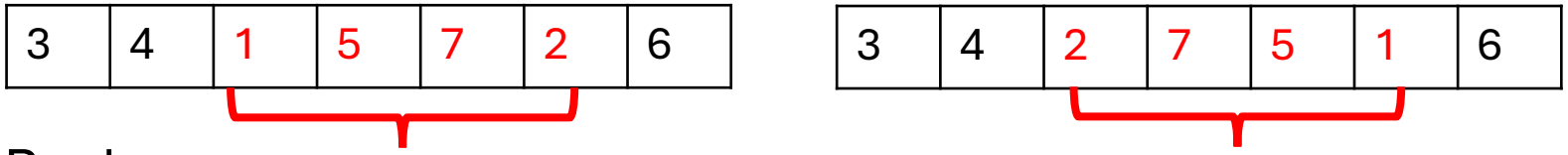
Chromosome A

Mutated

- Scramble




- Inversion




- Boundary-Replaces

# Termination

- After the reproduction steps we may decide to terminate if:
  - Fitness Threshold
  - Generational Limit
- The resultant output is the optimal solution.

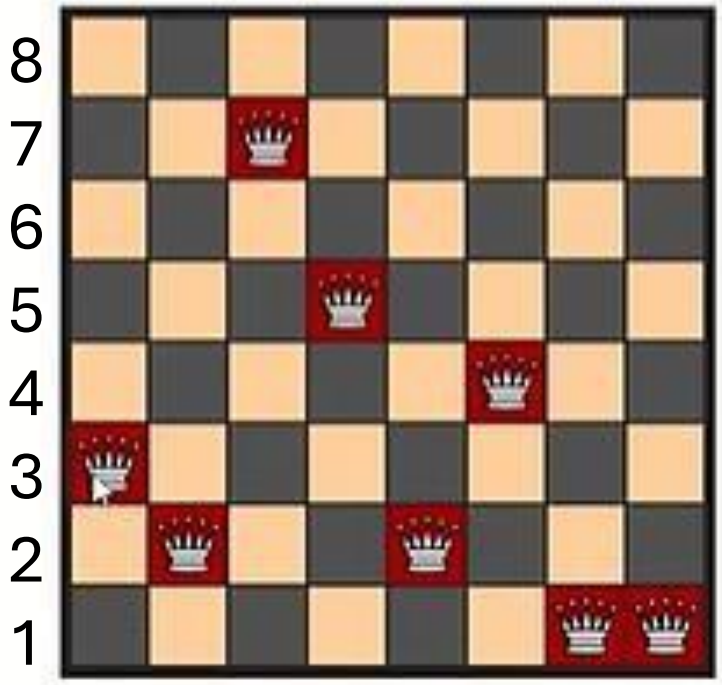
# GA

- Step 1: Representing individuals.
- Step 2: Generating a random initial Population
- Step 3: Apply a Fitness Function.
- Step 4: Selecting parents
- Step 5: Crossover of parents to produce new generation.
- Step 6: Mutation of new generation to bring diversity.
- Step 7: Repeat until solution is reached.

# 8 Queens Problem-GA
# Step 1: Representing individuals.

- Formulate an appropriate method to represent individuals of a population.
- Array.
- Index: Column.
- Value: Row.

| 3 | 2 | 7 | 5 | 2 | 4 | 1 | 1 |
|---|---|---|---|---|---|---|---|



# Step 2: Generating a random initial Population

- Generate random arrangements of 8 queens on a standard chess board.

A

| 3 | 2 | 7 | 5 | 2 | 4 | 1 | 1 |
|---|---|---|---|---|---|---|---|

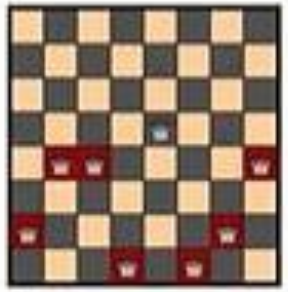
B

| 2 | 4 | 7 | 4 | 8 | 5 | 5 | 2 |
|---|---|---|---|---|---|---|---|

C

| 3 | 2 | 5 | 4 | 3 | 2 | 1 | 3 |
|---|---|---|---|---|---|---|---|

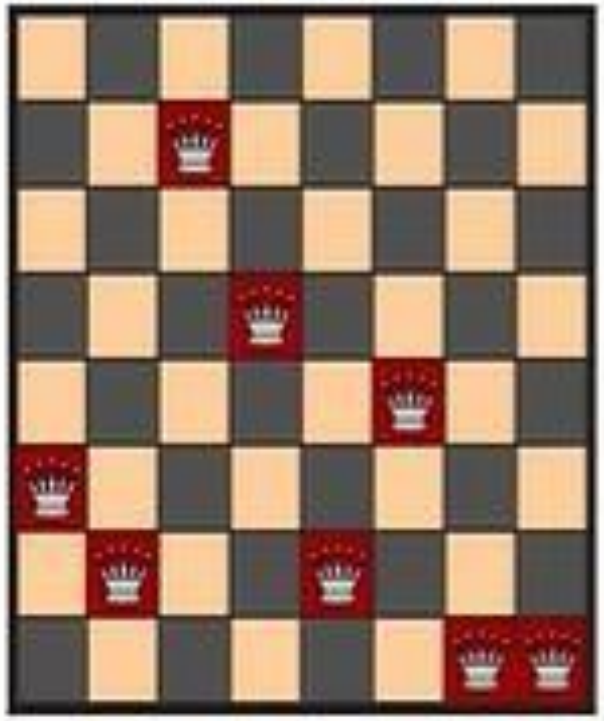
D



| 2 | 4 | 4 | 1 | 5 | 1 | 2 | 4 |
|---|---|---|---|---|---|---|---|

# Step 3: Apply a Fitness Function.

Individual



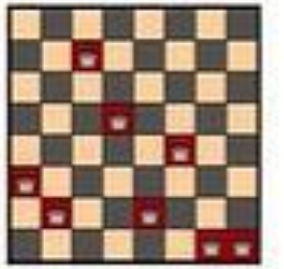
| 3 | 2 | 7 | 5 | 2 | 4 | 1 | 1 |
|---|---|---|---|---|---|---|---|

Fitness = No. of non attacking pairs

- Queen 1: 6
- Queen 2: 5
- Queen 3: 4
- Queen 4: 3
- Queen 5: 3
- Queen 6: 2
- Queen 7: 0
- Queen 8: 0

Total 23

# Step 3: Apply a Fitness Function.

| | A | B | C | D |
|---|---|---|---|---|
| Individuals |  3 2 7 5 2 4 1 1 |  2 4 7 4 8 5 5 2 |  3 2 5 4 3 2 1 3 |  2 4 4 1 5 1 2 4 |
| Fitness | 23 | 24 | 11 | 20 |
| Fitness % | 29% | 31% | 14% | 26% |

Total Fitness:78

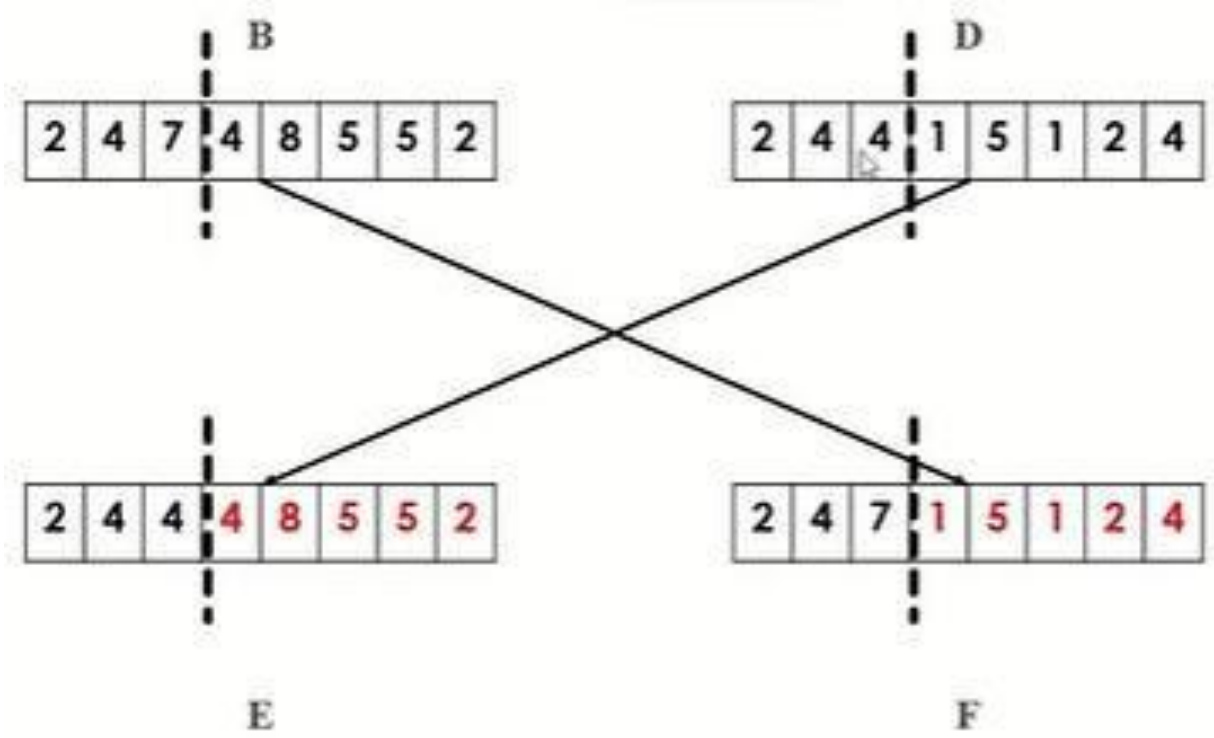
23/78=0.29

# Step 4: Selecting parents

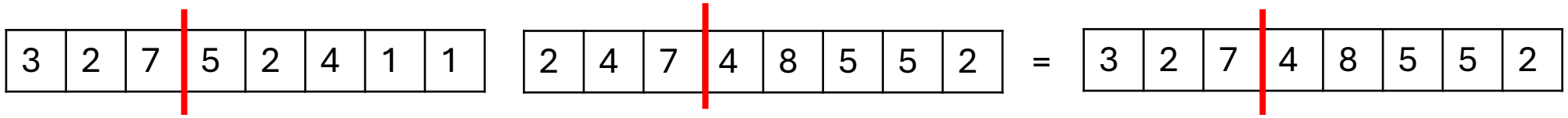
- Use any selection schemes
  - Tournament selection
  - Roulette wheel selection
  - Rank-based selection
  - Random selection

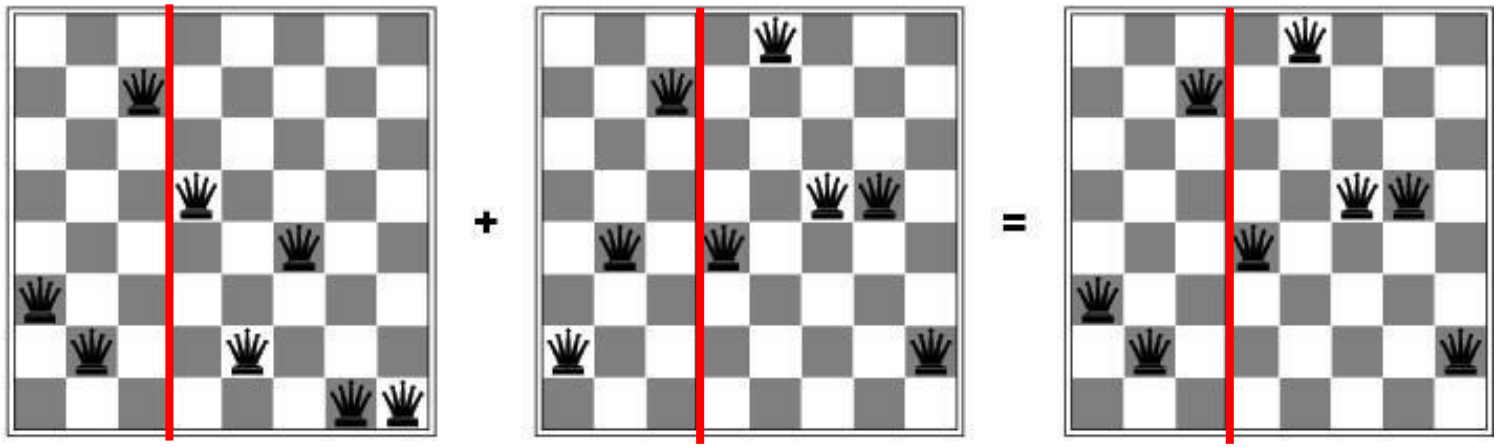
- Suppose B and D are selected

# Step 5: Crossover of parents to produce new generation.

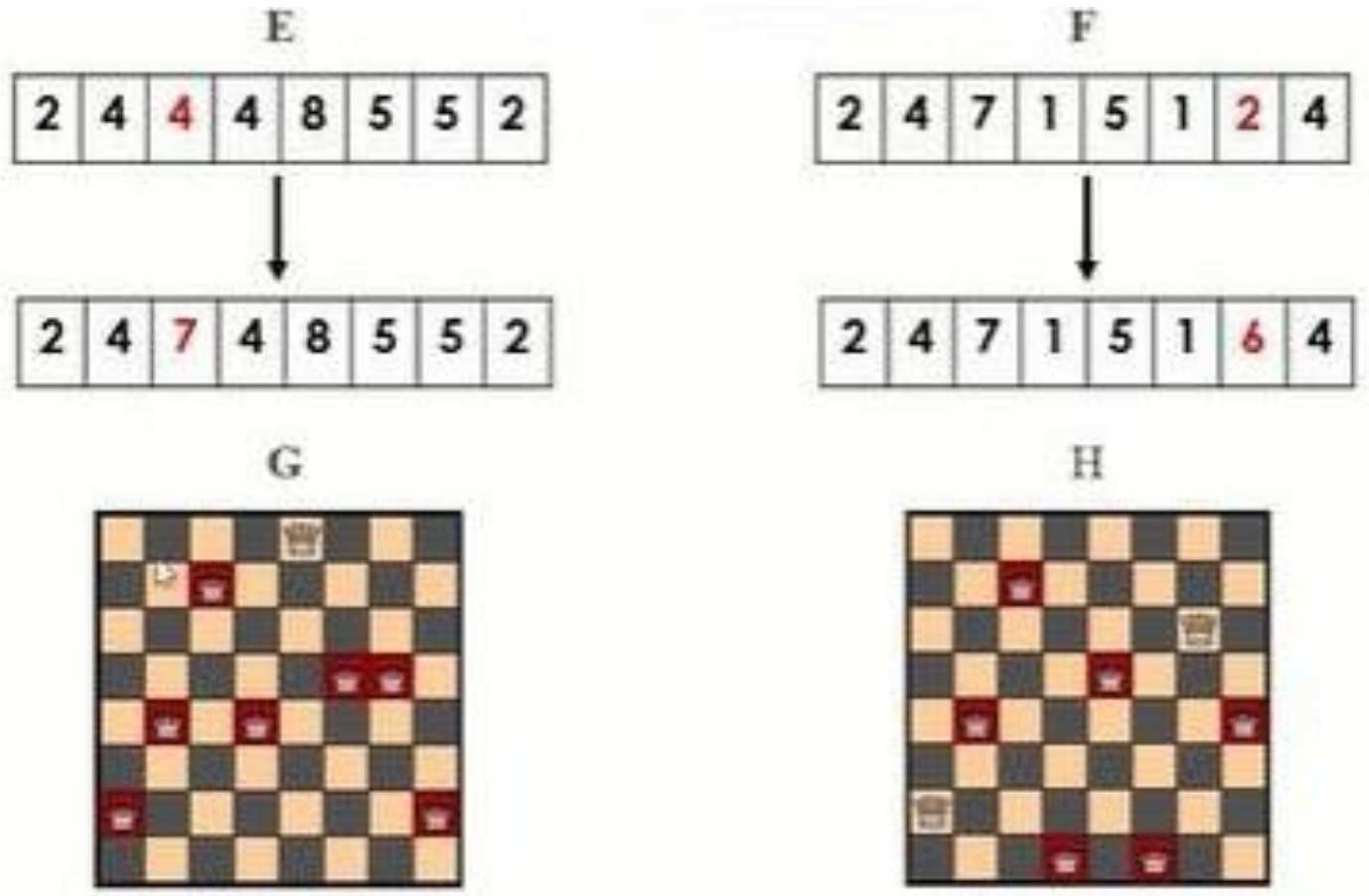


# Crossover operator



# Step 6: Mutation of new generation to bring diversity.



# Step 7: Repeat until solution is reached.

- All steps are repeated until best solution is reached.
- Best solution = Highest fitness score (28 in this case).

# Example: Knapsack Problem (Capacity=12kg)

| Item | Weight (kg) | Value |
|---|---|---|
| A | 5 | 10 |
| B | 3 | 5 |
| C | 7 | 9 |
| D | 2 | 8 |

Problem is that which item should be kept in the knapsack so as it will maximize knapsack value without breaking knapsack

# Example: Knapsack Problem (Capacity=12kg)

*0 represents absence of item
*1 represent the presence of item
State Space=$2^4$

## Population

| | A | B | C | D |
|---|---|---|---|---|
| C1 | 0 | 1 | 1 | 0 |
| C2 | 0 | 0 | 1 | 1 |
| C3 | 0 | 1 | 1 | 1 |
| C4 | 1 | 0 | 1 | 0 |

## Fitness

Fitness Function for 1:
Weight=3+ 7=10
Value=5+9=14

Fitness Function for 2:
Weight=7+ 2=9
Value=8+9=17

Fitness Function for 3:
Weight=3+ 7+2=12
Value=5+9+8=22

Fitness Function for 4:
Weight=5+ 7=12
Value=10+9=19

## Selection

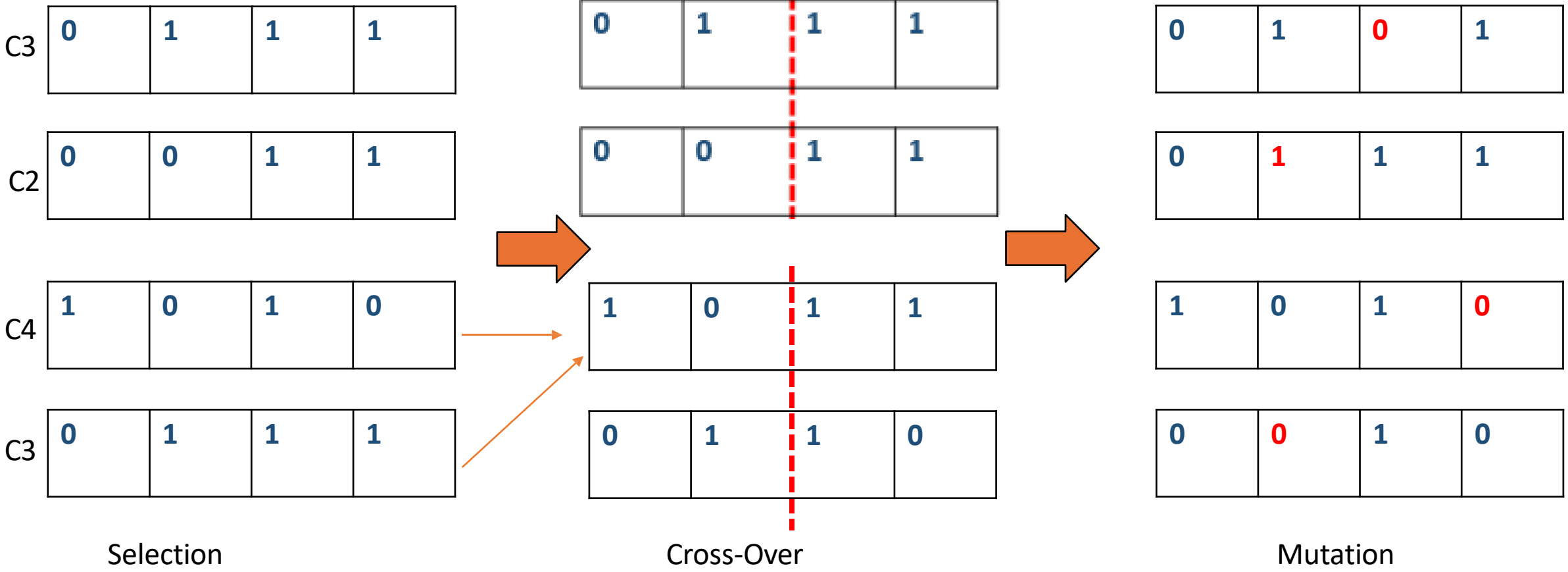
**Roulette Wheel selection:** the state that has the highest fitness value gets larger share of the wheel

| | | | | |
|---|---|---|---|---|
| 0 | 1 | 1 | 1 | C3 |
| 0 | 0 | 1 | 1 | C2 |
| 1 | 0 | 1 | 0 | C4 |
| 0 | 1 | 1 | 1 | C3 |

# Example: Knapsack Problem (Capacity=12kg)

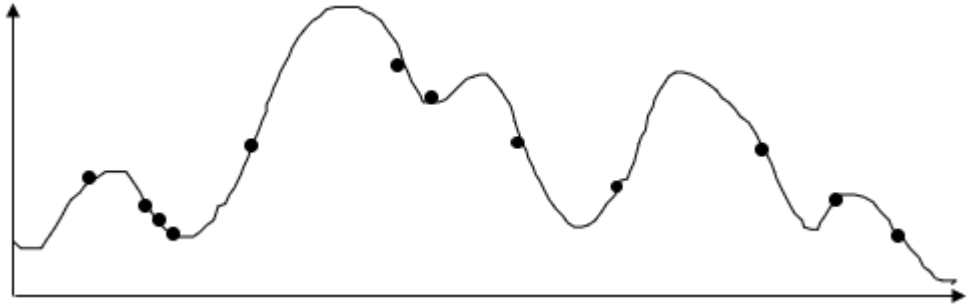
*0 represents absence of item
*1 represent the presence of item
State Space=$2^4$

Crossover:
Randomly select the position around which gene would be exchanged

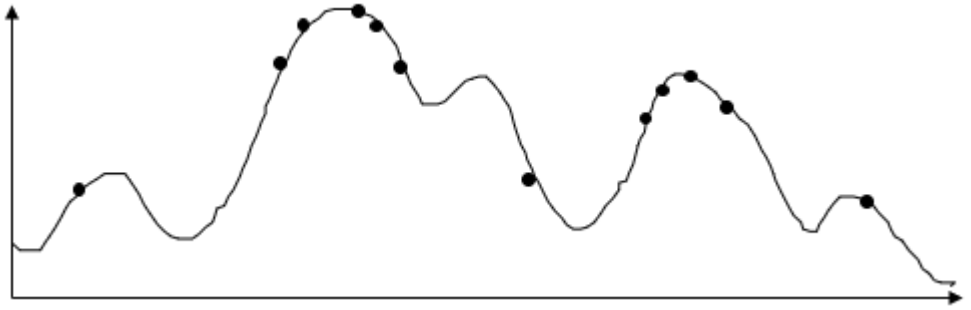


# An Abstract Example

- Distribution of Individuals in Generation 0



- Distribution of Individuals in Generation N



# Exercise

**Problem:**

Use a simple Genetic Algorithm to maximize the function:

$$f(x) = x^2$$

where $x$ is an integer in the range $[0, 15]$.

- Use the following settings:
  - Binary encoding (4 bits)
  - Population size = 4
  - Initial population:
    - 0101
    - 1100
    - 0011
    - 1010
  - Selection method: Roulette Wheel Selection
  - Crossover: Single-point crossover (after 2nd bit)
  - Mutation rate: 1 bit at a time

# Exercise

**Problem:**

Use a simple Genetic Algorithm to maximize the function:

$$f(x) = x^2$$

where $x$ is an integer in the range $[0, 15]$.

| Chromosome # | Chromosome | decimal (x) | Fitness =F(x) | Probability |
|---|---|---|---|---|
| A | 0101 | | | |
| B | 1100 | | | |
| C | 0011 | | | |
| D | 1010 | | | |
| | | | | |

# Exercise

**Problem:**

Use a simple Genetic Algorithm to maximize the function:

$$f(x) = x^2$$

where $x$ is an integer in the range $[0, 15]$.

| Chromosome # | Chromosome | decimal (x) | Fitness =F(x) | Probability |
|---|---|---|---|---|
| A | 0101 | 5 | 25 | 0.09 |
| B | 1100 | 12 | 144 | 0.52 |
| C | 0011 | 3 | 9 | 0.03 |
| D | 1010 | 10 | 100 | 0.36 |
| Sum | | | 278 | |

**Suppose B and C selected using RW**

# Exercise

**Problem:**

Use a simple Genetic Algorithm to maximize the function:

$$f(x) = x^2$$

where $x$ is an integer in the range $[0, 15]$.

| Chromosome # | Chromosome | decimal (x) | Fitness =F(x) | Probability |
|---|---|---|---|---|
| A | 0101 | 5 | 25 | |
| B | 1100 | 12 | 144 | |
| C | 0011 | 3 | 9 | |
| D | 1010 | 10 | 100 | |
| E | 1110 | 14 | 196 | |
| F | 0100 | 4 | 16 | |

**Suppose B and C selected using RW. They are crossed over to produce E s F**

# Exercise

**Problem:**

Use a simple Genetic Algorithm to maximize the function:

$$f(x) = x^2$$

where $x$ is an integer in the range $[0, 15]$.

| Chromosome # | Chromosome | decimal (x) | Fitness =F(x) | Probability |
|---|---|---|---|---|
| A | 0101 | 5 | 25 | |
| B | 1100 | 12 | 144 | |
| D | 1010 | 10 | 100 | |
| E | 1110 | 14 | 196 | |

# Types of Genetic Algorithms

- **Generational GA (Standard GA):**
  - Creates new offspring from the old population using genetic operators.
  - Entire new population is generated at once.
  - The new population completely replaces the old population for the next generation.
- **Incremental/Steady State GA:**
  - Introduces one new member into the population at a time.
  - Utilizes a replacement or deletion strategy to decide which existing member will be replaced by the new offspring.

# Some GA Application Types

| Domain | Application Types |
|---|---|
| **Control** | gas pipeline, pole balancing, missile evasion, pursuit |
| **Design** | semiconductor layout, aircraft design, keyboard configuration, communication networks |
| **Scheduling** | manufacturing, facility scheduling, resource allocation |
| **Robotics** | trajectory planning |
| **Machine Learning** | designing neural networks, improving classification algorithms, classifier systems |
| **Game Playing** | checkers, prisoner's dilemma |
| **Combinatorial Optimization** | travelling salesman, routing, bin packing, graph colouring and partitioning |

# Applications: Genetic algorithms

- DNA Analysis
- Image Processing
- Vehicle Routing
- Neural Network

### Solving Difficult Problems

GAs prove to be an efficient tool to provide **usable near-optimal solutions** in a short amount of time.

### Failure of Gradient Based Methods

In most real-world situations we have a very complex problem as it suffer from an inherent tendency of getting stuck at the local optima as shown here.

### Getting a Good Solution Fast

Some difficult problems like the Travelling Salesperson Problem (TSP), have real-world applications like path finding and VLSI Design.

# Advantages of Genetic Algorithms

- Modularity
- Robust in Noisy Environments
- Multi-Objective Optimization
- Improves Over Time
- Parallelism

# Issues for GA Practitioners

1. **Implementation Challenges**:
   Choosing representation, population size, mutation rate, and other parameters.
2. **Selection and Deletion Policies**:
   Deciding how individuals are selected and replaced in each generation.
3. **Crossover and Mutation Operators**:
   Selecting appropriate operators to generate better offspring.
4. **Performance and Scalability**:
   Ensuring the algorithm performs efficiently as the problem size increases.
5. **Evaluation Function**:
   The solution quality heavily depends on the evaluation function, which can be the most challenging part to design.

# Performance Measures for Single-Objective Optimization Algorithms

- **Memory Usage**
  - **Resource Utilization**: The amount of memory and computational resources required during execution.
- **Termination Criteria**
  - **Stopping Conditions**: Evaluating whether the algorithm terminates effectively based on predefined criteria, such as reaching a specific accuracy, maximum iterations, or time limit.

- These measures help in comparing different algorithms and selecting the most suitable one for a given optimization problem.